

Lecture 03 - Geometric Transformations

This lecture explains how geometry moves between coordinate systems. Almost every later topic assumes that you can track where a point or vector currently lives.

Source chunk	course_text_parts/03_lectures/03_geometric-transformations.txt
Extracted slide pages	54
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Untitled slide
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Untitled slide. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Untitled slide' into a causal sentence instead of repeating the slide title?

Page 2	Transforming 3D Models
Source cue	- 3D models need to be positioned and oriented in the virtual world / (sometimes scaling and other modifications are necessary) / - These modifications are performed by applying transformations to the / vertices describing a 3D model / - Transformations are applied through
Commentary	This slide is about Transforming 3D Models. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - 3D models need to be positioned and oriented in the virtual world / (sometimes scaling and other modifications are necessary) / - These modifications are performed by applying transformations to the / vertices describing a 3D model / - Transformations are applied through
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.

Check yourself	Can you state the coordinate space before and after 'Transforming 3D Models'?
Page 3	3.1 Mathematical Foundations
Source cue	3.2 Affine Transformations / 3.3 Composition of Transformations / 3.4 Coordinate System Change / 3.5 Transformations in OpenGL
Commentary	This slide is about 3.1 Mathematical Foundations. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: 3.2 Affine Transformations / 3.3 Composition of Transformations / 3.4 Coordinate System Change / 3.5 Transformations in OpenGL
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '3.1 Mathematical Foundations'?
Page 4	3.1 Mathematical Foundations
Source cue	Vector and Matrix Calculus
Commentary	This slide is about 3.1 Mathematical Foundations. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Vector and Matrix Calculus
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '3.1 Mathematical Foundations'?
Page 5	Vertices as Vectors
Source cue	- Vectors are used to represent geometric objects (coordinates) / - Vectors have real numbers as components, so they are elements of $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^4$ / - In computer graphics mostly $\mathbb{R}^3$ or $\mathbb{R}^4$ vectors used / - Vertices of a polygon are represented as column vectors / $x \in \mathbb{R}^n$
Commentary	This slide is about Vertices as Vectors. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Vectors are used to represent geometric objects (coordinates) / - Vectors have real numbers as components, so they are elements of $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^4$ / - In computer graphics mostly $\mathbb{R}^3$ or $\mathbb{R}^4$ vectors used / - Vertices of a polygon are represented as column vectors / $x \in \mathbb{R}^n$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Vertices as Vectors'?
Page 6	Transformations as Matrices
Source cue	- Matrices are used to represent transformations / - In computer graphics mainly 4×4 square matrices are used / $M \in \mathbb{R}^{n \times n}$ / $M \in \mathbb{R}^{n \times n}$ / $M =$

Commentary	This slide is about Transformations as Matrices. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrices are used to represent transformations / - In computer graphics mainly 4×4 square matrices are used / $\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} / \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} =$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Transformations as Matrices'?
Page 7	Matrix Vector Multiplication
Source cue	- Since vectors describe vertices and matrices describe transformations, / transformations can be applied through matrix vector multiplication / - Input: $n \times m$ matrix M and m vector v / - Result: n vector u , whereby is v transformed through M / - Example for 4×4 matrix M and 4 vector v :
Commentary	This slide is about Matrix Vector Multiplication. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Since vectors describe vertices and matrices describe transformations, / transformations can be applied through matrix vector multiplication / - Input: $n \times m$ matrix M and m vector v / - Result: n vector u , whereby is v transformed through M / - Example for 4×4 matrix M and 4 vector v :
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Matrix Vector Multiplication'?
Page 8	3.2 Affine Transformations
Source cue	Translation, Rotation, Scaling, and more
Commentary	This slide is about 3.2 Affine Transformations. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Translation, Rotation, Scaling, and more
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '3.2 Affine Transformations'?
Page 9	Translations – Moving 3D Models
Source cue	- Matrix representation of a translation / $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} =$ / $x \ y \ z$
Commentary	This slide is about Translations – Moving 3D Models. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrix representation of a translation / $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} =$ / $x \ y \ z$

Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Translations – Moving 3D Models'?
Page 10	Translation Properties
Source cue	- Lengths and angles are preserved / - Identity: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (identity matrix) / - Changing order of translations: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} / x y z x y z x y z x y z$
Commentary	This slide is about Translation Properties. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Lengths and angles are preserved / - Identity: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (identity matrix) / - Changing order of translations: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} / x y z x y z x y z x y z$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Translation Properties'?
Page 11	Scaling – Resizing 3D Models
Source cue	- Matrix representation of a scaling / $\begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} = \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix}, s, s, s = / x y z$
Commentary	This slide is about Scaling – Resizing 3D Models. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrix representation of a scaling / $\begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} = \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix}, s, s, s = / x y z$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Scaling – Resizing 3D Models'?
Page 12	Scaling Properties
Source cue	- Lengths are not preserved, angles only for uniform scalings / - Identity: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (identity matrix) / - Changing order of scalings: $\begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} \cdot \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} = \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} \cdot \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} / x y z x y z x y z x y z$
Commentary	This slide is about Scaling Properties. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Lengths are not preserved, angles only for uniform scalings / - Identity: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (identity matrix) / - Changing order of scalings: $\begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} \cdot \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} = \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} \cdot \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix} / x y z x y z x y z x y z$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Scaling Properties'?

Page 13	Rotation – Changing Orientation of 3D Models 1/2
Source cue	- Matrix representation of a rotation around x-, y- and z-axis / $1 \ 0 \ 0 \ 0 / 0 \ 0 \cos \theta \ -\sin \theta \ 0 / \sin \theta \ \cos \theta \ 0 = / 0 \ \sin \theta \ \sin \theta \ 0$
Commentary	This slide is about Rotation – Changing Orientation of 3D Models 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrix representation of a rotation around x-, y- and z-axis / $1 \ 0 \ 0 \ 0 / 0 \ 0 \cos \theta \ -\sin \theta \ 0 / \sin \theta \ \cos \theta \ 0 = / 0 \ \sin \theta \ \sin \theta \ 0$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Rotation – Changing Orientation of 3D Models 1/2'?
Page 14	Rotation – Changing Orientation of 3D Models 2/2
Source cue	- Rotations are made in the right-handed coordinate system so that when we / look along a positive axis to the origin, a 90° rotation about that axis / interleaves the positive axes as follows / - Rotation around x-axis transforms +y to +z / - Rotation around y-axis transforms +z to +x
Commentary	This slide is about Rotation – Changing Orientation of 3D Models 2/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Rotations are made in the right-handed coordinate system so that when we / look along a positive axis to the origin, a 90° rotation about that axis / interleaves the positive axes as follows / - Rotation around x-axis transforms +y to +z / - Rotation around y-axis transforms +z to +x
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Rotation – Changing Orientation of 3D Models 2/2'?
Page 15	Rotation Properties
Source cue	- Lengths and angles are preserved / - Identity: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$ (identity matrix) / x y z / - Changing order of rotations around same axis: / - $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta + \phi) & \sin(\theta + \phi) \\ 0 & -\sin(\theta + \phi) & \cos(\theta + \phi) \end{pmatrix}$
Commentary	This slide is about Rotation Properties. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Lengths and angles are preserved / - Identity: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$ (identity matrix) / x y z / - Changing order of rotations around same axis: / - $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta + \phi) & \sin(\theta + \phi) \\ 0 & -\sin(\theta + \phi) & \cos(\theta + \phi) \end{pmatrix}$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Rotation Properties'?
Page 16	Mirroring – Reflecting 3D Models on a Plane

Source cue	- Matrix representation of mirroring along x-y plane / $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}$ =
Commentary	This slide is about Mirroring – Reflecting 3D Models on a Plane. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrix representation of mirroring along x-y plane / $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}$ =
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Mirroring – Reflecting 3D Models on a Plane'?

Page 17	Shearing – Skewing 3D Models
Source cue	- Matrix representation of a shearing / $\begin{pmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ / xy xz / $\begin{pmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ / yx yz
Commentary	This slide is about Shearing – Skewing 3D Models. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrix representation of a shearing / $\begin{pmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ / xy xz / $\begin{pmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ / yx yz
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Shearing – Skewing 3D Models'?

Page 18	3.3 Composition of Transformations
Source cue	Concatenation of several transformation steps
Commentary	This slide is about 3.3 Composition of Transformations. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Concatenation of several transformation steps
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '3.3 Composition of Transformations'?

Page 19	Composition of Transformations
Source cue	- Composition of transformations $\mathbf{T}$ is done / by multiplying the respective transformation matrices / - $(\mathbf{T}_1 \cdot \mathbf{T}_2 \cdot \dots \cdot \mathbf{T}_n) \cdot \mathbf{P} = (\mathbf{T}_1 \cdot (\mathbf{T}_2 \cdot (\dots (\mathbf{T}_n \cdot (\mathbf{T}_1 \cdot \mathbf{P}))))$ / $\begin{pmatrix} 1 & 2 & n-1 & n \\ 1 & 2 & n-1 & n \end{pmatrix}$ / - Interpretation: apply first transformation $\mathbf{T}_1$, then $\mathbf{T}_2$, ..., and finally $\mathbf{T}_n$
Commentary	This slide is about Composition of Transformations. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Composition of transformations $\mathbf{T}$ is done / by multiplying the respective transformation matrices / - $(\mathbf{T}_1 \cdot \mathbf{T}_2 \cdot \dots \cdot \mathbf{T}_n) \cdot \mathbf{P} = (\mathbf{T}_1 \cdot (\mathbf{T}_2 \cdot (\dots (\mathbf{T}_n \cdot (\mathbf{T}_1 \cdot \mathbf{P}))))$ / $\begin{pmatrix} 1 & 2 & n-1 & n \\ 1 & 2 & n-1 & n \end{pmatrix}$ / - Interpretation: apply first transformation $\mathbf{T}_1$, then $\mathbf{T}_2$, ..., and finally $\mathbf{T}_n$

Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Composition of Transformations'?
Page 20	Reference Points 1/2
Source cue	- Rotation and scaling happen in relation to origin (reference point)
Commentary	This slide is about Reference Points 1/2. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Rotation and scaling happen in relation to origin (reference point)
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Reference Points 1/2' into a causal sentence instead of repeating the slide title?
Page 21	Reference Points 2/2
Source cue	- In order to rotate (or scale) 3D models at an arbitrary reference point, / a sequence of transformations is necessary / - Move the 3D model from reference point ■ to the origin ■ / - Rotate (or scale) the 3D model / - Move the 3D model back to ■
Commentary	This slide is about Reference Points 2/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - In order to rotate (or scale) 3D models at an arbitrary reference point, / a sequence of transformations is necessary / - Move the 3D model from reference point ■ to the origin ■ / - Rotate (or scale) the 3D model / - Move the 3D model back to ■
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Reference Points 2/2'?
Page 22	Rotation Around Arbitrary Axis
Source cue	- Rotation around any axis / - Axis given by center of rotation ■ and direction ■ = (■, ■, ■) / x y z / - Rotation by ■ degrees / - Construction of transformation as transformation sequence
Commentary	This slide is about Rotation Around Arbitrary Axis. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Rotation around any axis / - Axis given by center of rotation ■ and direction ■ = (■, ■, ■) / x y z / - Rotation by ■ degrees / - Construction of transformation as transformation sequence
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Rotation Around Arbitrary Axis'?

Page 23	Reminder: Vector Calculus
Source cue	- Length of a vector is expressed as follows / $ \mathbf{v} = \sqrt{v_x^2 + v_y^2 + v_z^2}$ / - Vectors of length $ \mathbf{v} = 1$ are normalized and called unit vectors / - Arbitrary vectors can be normalized as follows
Commentary	This slide is about Reminder: Vector Calculus. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Length of a vector is expressed as follows / $ \mathbf{v} = \sqrt{v_x^2 + v_y^2 + v_z^2}$ / - Vectors of length $ \mathbf{v} = 1$ are normalized and called unit vectors / - Arbitrary vectors can be normalized as follows
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reminder: Vector Calculus'?
Page 24	Reminder: Cross Product (= vector product)
Source cue	- Given two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$ / - Cross product $\mathbf{a} \times \mathbf{b}$ of $\mathbf{a}$ and $\mathbf{b}$ is a vector defined as follows / $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y) \mathbf{e}_1 + (a_z b_x - a_x b_z) \mathbf{e}_2 + (a_x b_y - a_y b_x) \mathbf{e}_3$
Commentary	This slide is about Reminder: Cross Product (= vector product). Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Given two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$ / - Cross product $\mathbf{a} \times \mathbf{b}$ of $\mathbf{a}$ and $\mathbf{b}$ is a vector defined as follows / $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y) \mathbf{e}_1 + (a_z b_x - a_x b_z) \mathbf{e}_2 + (a_x b_y - a_y b_x) \mathbf{e}_3$
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Reminder: Cross Product (= vector product)' into a causal sentence instead of repeating the slide title?
Page 25	Example – Transforming Directed Line Segment 1/7
Source cue	- Example transformation task / - Given line segments $\mathbf{p}_1, \mathbf{p}_2$ in $\mathbb{R}^3$ / - Transform it into the yz-plane so that $\mathbf{p}_1$ lies on the z-axis / 1 2
Commentary	This slide is about Example – Transforming Directed Line Segment 1/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Example transformation task / - Given line segments $\mathbf{p}_1, \mathbf{p}_2$ in $\mathbb{R}^3$ / - Transform it into the yz-plane so that $\mathbf{p}_1$ lies on the z-axis / 1 2
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 1/7'?
Page 26	Example – Transforming Directed Line Segment 2/7
Source cue	- Approach 1: Determine rotation angles for rotation sequence / - Move $\mathbf{p}_1 = x, y, z$ into the origin / 1 1 1 1 / - Rotate around the y-axis so that $\mathbf{p}_1$ lies in the yz-plane / 1 2

Commentary	This slide is about Example – Transforming Directed Line Segment 2/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Approach 1: Determine rotation angles for rotation sequence / - Move $\mathbf{x}, y, z$ into the origin / $1 \ 1 \ 1 \ 1$ / - Rotate around the y-axis so that $\mathbf{x} \ \mathbf{y}$ lies in the yz-plane / $1 \ 2$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 2/7'?
Page 27	Example – Transforming Directed Line Segment 3/7
Source cue	- Step 2 – Rotation of $\mathbf{x}' \ \mathbf{y}'$ around y-axis into yz-plane / $1 \ 2$ / - Rotation $\mathbf{x}'$ ($\mathbf{x}' - 90^\circ$) / - Rotation angle – $(90^\circ - \mathbf{x}') = \mathbf{x}' - 90^\circ$ / - Matrix ingredients
Commentary	This slide is about Example – Transforming Directed Line Segment 3/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Step 2 – Rotation of $\mathbf{x}' \ \mathbf{y}'$ around y-axis into yz-plane / $1 \ 2$ / - Rotation $\mathbf{x}'$ ($\mathbf{x}' - 90^\circ$) / - Rotation angle – $(90^\circ - \mathbf{x}') = \mathbf{x}' - 90^\circ$ / - Matrix ingredients
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 3/7'?
Page 28	Example – Transforming Directed Line Segment 4/7
Source cue	- Step 3 – Rotation of $\mathbf{x}'' \ \mathbf{y}''$ around x-axis onto z-axis / $1 \ 2$ / - Rotation $\mathbf{x}''$ (ϕ) / - Matrix ingredients / z''
Commentary	This slide is about Example – Transforming Directed Line Segment 4/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Step 3 – Rotation of $\mathbf{x}'' \ \mathbf{y}''$ around x-axis onto z-axis / $1 \ 2$ / - Rotation $\mathbf{x}''$ (ϕ) / - Matrix ingredients / z''
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 4/7'?
Page 29	Example – Transforming Directed Line Segment 5/7
Source cue	- Step 4 – Rotation of $\mathbf{x}''' \ \mathbf{y}'''$ into yz-plane / $1 \ 3$ / - Rotation $\mathbf{x}'''$ (α) / $y \ \mathbf{x}''' \ \mathbf{z}'''$ / - Matrix ingredients
Commentary	This slide is about Example – Transforming Directed Line Segment 5/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Step 4 – Rotation of $\mathbf{x}''' \ \mathbf{y}'''$ into yz-plane / $1 \ 3$ / - Rotation $\mathbf{x}'''$ (α) / $y \ \mathbf{x}''' \ \mathbf{z}'''$ / - Matrix ingredients
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.

Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 5/7'?
Page 30	Example – Transforming Directed Line Segment 6/7
Source cue	- Approach 2: Construct orthogonal matrix which obeys following properties / - Each row and column corresponds to a unit length vector / - Row and column vectors are orthogonal to each other / - Each row vector rotates on one main axis / (■ on x-axis, ■ on y-axis, ■ on z-axis)
Commentary	This slide is about Example – Transforming Directed Line Segment 6/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Approach 2: Construct orthogonal matrix which obeys following properties / - Each row and column corresponds to a unit length vector / - Row and column vectors are orthogonal to each other / - Each row vector rotates on one main axis / (■ on x-axis, ■ on y-axis, ■ on z-axis)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 6/7'?
Page 31	Example – Transforming Directed Line Segment 7/7
Source cue	- Consideration 2: Formulate ■ as unit vector / orthogonal to the plane spanned by ■ , ■ , and ■ / $1\ 2\ 3 / \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} \times \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} / - \blacksquare = \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} = 1\ 3\ 1\ 2$
Commentary	This slide is about Example – Transforming Directed Line Segment 7/7. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Consideration 2: Formulate ■ as unit vector / orthogonal to the plane spanned by ■ , ■ , and ■ / $1\ 2\ 3 / \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} \times \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} / - \blacksquare = \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} = 1\ 3\ 1\ 2$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Transforming Directed Line Segment 7/7'?
Page 32	Example – Direction of Flight Transformation 1/3
Source cue	- Example transformation task / - Given 3D model centered in origin of x -, y -, z -coordinate system / ■ ■ ■ / - Transform it such that / - it is centered in arbitrary point ■ , and
Commentary	This slide is about Example – Direction of Flight Transformation 1/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Example transformation task / - Given 3D model centered in origin of x -, y -, z -coordinate system / ■ ■ ■ / - Transform it such that / - it is centered in arbitrary point ■ , and
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.

Check yourself	Can you state the coordinate space before and after 'Example – Direction of Flight Transformation 1/3'?
Page 33	Example – Direction of Flight Transformation 2/3
Source cue	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1x \ 2x \ 3x / - \text{Step 1} - \text{create orthogonal rotation matrix } \mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1y \ 2y \ 3y / \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Commentary	This slide is about Example – Direction of Flight Transformation 2/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1x \ 2x \ 3x / - \text{Step 1} - \text{create orthogonal rotation matrix } \mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1y \ 2y \ 3y / \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Direction of Flight Transformation 2/3'?
Page 34	Example – Direction of Flight Transformation 3/3
Source cue	- Transformation $\mathbf{T}$ is described by a translation and an orthogonal matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1x \ 2x \ 3x / \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1y \ 2y \ 3y$
Commentary	This slide is about Example – Direction of Flight Transformation 3/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Transformation $\mathbf{T}$ is described by a translation and an orthogonal matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1x \ 2x \ 3x / \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} / 1y \ 2y \ 3y$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example – Direction of Flight Transformation 3/3'?
Page 35	3.4 Coordinate System Change
Source cue	Transformation as change of frame of reference
Commentary	This slide is about 3.4 Coordinate System Change. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Transformation as change of frame of reference
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '3.4 Coordinate System Change'?
Page 36	Coordinate System Change
Source cue	- Alternative view on transformations: / - transformations change the coordinate system of 3D model / Example 1 Example 2

Commentary	This slide is about Coordinate System Change. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Alternative view on transformations: / - transformations change the coordinate system of 3D model / Example 1 Example 2
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Coordinate System Change'?
Page 37	Local Coordinate Systems
Source cue	- Application in OpenGL / - Individual 3D models of a scene have their own, local coordinate system / - Local coordinate systems of the 3D models must be changed / into a common coordinate system, the world coordinate system / - Transformation between coordinate systems (CS)
Commentary	This slide is about Local Coordinate Systems. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Application in OpenGL / - Individual 3D models of a scene have their own, local coordinate system / - Local coordinate systems of the 3D models must be changed / into a common coordinate system, the world coordinate system / - Transformation between coordinate systems (CS)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Local Coordinate Systems'?
Page 38	Example 1 – Coordinate System Change 1/2
Source cue	$x(1)$ / - Example transformation task $y(5)$ / - Rotation and scaling of an object around its center $\blacksquare$ / $y(1)$ / and subsequent translation to a point $\blacksquare$
Commentary	This slide is about Example 1 – Coordinate System Change 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: $x(1)$ / - Example transformation task $y(5)$ / - Rotation and scaling of an object around its center $\blacksquare$ / $y(1)$ / and subsequent translation to a point $\blacksquare$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example 1 – Coordinate System Change 1/2'?
Page 39	Example 1 – Coordinate System Change 2/2
Source cue	$y(5)$ / $y(2)$ / $y(3)$ / $x(1)$ / $x(2)$ / $x(3)$ / 3D Model Transformation / $y(5)$ / $y(2)$ / $y(3)$
Commentary	This slide is about Example 1 – Coordinate System Change 2/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: $y(5)$ / $y(2)$ / $y(3)$ / $x(1)$ / $x(2)$ / $x(3)$ / 3D Model Transformation / $y(5)$ / $y(2)$ / $y(3)$

Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example 1 – Coordinate System Change 2/2'?
Page 40	Example 2 – Hierarchical Coordinate Systems 1/2
Source cue	- To model complex scenes, a hierarchy of coordinate systems is helpful / - Example: modeling of a tricycle / - World-coordinate system / - Tricycle-coordinate system / - Wheel-coordinate system
Commentary	This slide is about Example 2 – Hierarchical Coordinate Systems 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - To model complex scenes, a hierarchy of coordinate systems is helpful / - Example: modeling of a tricycle / - World-coordinate system / - Tricycle-coordinate system / - Wheel-coordinate system
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example 2 – Hierarchical Coordinate Systems 1/2'?
Page 41	Example 2 – Hierarchical Coordinate Systems 1/2
Source cue	- Scene graphs are used to represent hierarchical coordinate systems / and the required transformations / Tricycle Translation T / Steering Rotation $R(\alpha)$ / y-tr
Commentary	This slide is about Example 2 – Hierarchical Coordinate Systems 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Scene graphs are used to represent hierarchical coordinate systems / and the required transformations / Tricycle Translation T / Steering Rotation $R(\alpha)$ / y-tr
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Example 2 – Hierarchical Coordinate Systems 1/2'?
Page 42	3.5 Transformations in OpenGL
Source cue	Implementation with GLM und GLSL
Commentary	This slide is about 3.5 Transformations in OpenGL. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Implementation with GLM und GLSL
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '3.5 Transformations in OpenGL'?
Page 43	Geometric Transformations in OpenGL

Source cue	- Supported coordinate systems / - Floating-point coordinate system for 2D and 3D / - Integer coordinate system (2D) as a special case / - Representation of vectors / - 2D vector by one-dimensional array with 2 elements
Commentary	This slide is about Geometric Transformations in OpenGL. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Supported coordinate systems / - Floating-point coordinate system for 2D and 3D / - Integer coordinate system (2D) as a special case / - Representation of vectors / - 2D vector by one-dimensional array with 2 elements
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Geometric Transformations in OpenGL'?

Page 44	OpenGL Transformation Pipeline
Source cue	- Model-View matrix / - Transformation matrix for model space related transformations / - Additionally contains the transformation of the scene / into the camera coordinate system (view transformation, not projection!) / - Transformation pipeline
Commentary	This slide is about OpenGL Transformation Pipeline. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Model-View matrix / - Transformation matrix for model space related transformations / - Additionally contains the transformation of the scene / into the camera coordinate system (view transformation, not projection!) / - Transformation pipeline
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Transformation Pipeline'?

Page 45	Example – Translation and Scaling
Source cue	- Drawing cylinder objects / - drawCylinder draws a cylinder centered at the origin, / with respect to the y-axis, with a given radius and height / - By translation this "unit cylinder" can be positioned / - By scaling this "unit cylinder" can be resized
Commentary	This slide is about Example – Translation and Scaling. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Drawing cylinder objects / - drawCylinder draws a cylinder centered at the origin, / with respect to the y-axis, with a given radius and height / - By translation this "unit cylinder" can be positioned / - By scaling this "unit cylinder" can be resized
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Example – Translation and Scaling' into a causal sentence instead of repeating the slide title?

Page 46	Matrix Specification
Source cue	- Matrices are often specified in form of a C array / void draw(...) { / ... / GLfloat M[4][4]; / M[0][0] =;
Commentary	This slide is about Matrix Specification. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Matrices are often specified in form of a C array / void draw(...) { / ... / GLfloat M[4][4]; / M[0][0] =;
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Matrix Specification'?
Page 47	Example – Portico Modeling 1/5
Source cue	- Columned hall consists of floor space, H·0.1 / top surface and columns / - Columns are evenly distributed / along the edge of the round floor surface H / - Total height of a column is ■,
Commentary	This slide is about Example – Portico Modeling 1/5. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Columned hall consists of floor space, H·0.1 / top surface and columns / - Columns are evenly distributed / along the edge of the round floor surface H / - Total height of a column is ■,
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Example – Portico Modeling 1/5' into a causal sentence instead of repeating the slide title?
Page 48	Example – Portico Modeling 2/5
Source cue	- Drawing a cylinder closed at top and bottom / - Each column should "stand" in the y-axis / - Each column should have a lid surface at the top and bottom / - Disk primitive is embedded in the xy-plane / - Cylinder primitive consists only of lateral surface
Commentary	This slide is about Example – Portico Modeling 2/5. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Drawing a cylinder closed at top and bottom / - Each column should "stand" in the y-axis / - Each column should have a lid surface at the top and bottom / - Disk primitive is embedded in the xy-plane / - Cylinder primitive consists only of lateral surface
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Example – Portico Modeling 2/5' into a causal sentence instead of repeating the slide title?
Page 49	Example – Portico Modeling 3/5

Source cue	- Drawing a single column / - Composition of 5 cylinder objects / - Rotation into y-axis to match orientation / - Positioning of the parts by translation along the y-axis / - Incremental translation around the partial column height
Commentary	This slide is about Example – Portico Modeling 3/5. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Drawing a single column / - Composition of 5 cylinder objects / - Rotation into y-axis to match orientation / - Positioning of the parts by translation along the y-axis / - Incremental translation around the partial column height
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Example – Portico Modeling 3/5' into a causal sentence instead of repeating the slide title?
Page 50	Example – Portico Modeling 4/5
Source cue	- Drawing the group of columns / - Columns are evenly distributed on a circle with radius ■ / - Number ■, radius ■ of the columns as well as floor radius ■ adjustable
Commentary	This slide is about Example – Portico Modeling 4/5. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Drawing the group of columns / - Columns are evenly distributed on a circle with radius ■ / - Number ■, radius ■ of the columns as well as floor radius ■ adjustable
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Example – Portico Modeling 4/5' into a causal sentence instead of repeating the slide title?
Page 51	Example – Portico Modeling 5/5
Source cue	- Drawing the hall / - Use cylinder primitive as floor area / - Use cone primitive as a roof
Commentary	This slide is about Example – Portico Modeling 5/5. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Drawing the hall / - Use cylinder primitive as floor area / - Use cone primitive as a roof
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Example – Portico Modeling 5/5' into a causal sentence instead of repeating the slide title?
Page 52	by applying affine transformations to their vertices

Source cue	- 3D models can be positioned, resized and oriented in the 3D world / by applying affine transformations to their vertices / - An affine transformation is represented by a 4x4 matrix / - Affine transformations can be more efficiently applied / when concatenated to a single matrix through matrix multiplication
Commentary	This slide is about by applying affine transformations to their vertices. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - 3D models can be positioned, resized and oriented in the 3D world / by applying affine transformations to their vertices / - An affine transformation is represented by a 4x4 matrix / - Affine transformations can be more efficiently applied / when concatenated to a single matrix through matrix multiplication
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'by applying affine transformations to their vertices'?

Page 53	Literature and other sources used in this chapter
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Literature and other sources used in this chapter. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?

Page 54	(3rd Edition), Addison-Wesley 2013. (Chapters 10 & 11)
Source cue	- Text Books / - J. Foley, A. van Dam, S. Feiner: Computer Graphics: Principles and Practice / (3rd Edition), Addison-Wesley 2013. (Chapters 10 & 11) / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer
Commentary	This slide is about (3rd Edition), Addison-Wesley 2013. (Chapters 10 & 11). Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Text Books / - J. Foley, A. van Dam, S. Feiner: Computer Graphics: Principles and Practice / (3rd Edition), Addison-Wesley 2013. (Chapters 10 & 11) / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '(3rd Edition), Addison-Wesley 2013. (Chapters 10 & 11)' into a causal sentence instead of repeating the slide title?

03.1 Mathematical Foundations

Source location: Section 3.1

Commentary

The lecture uses vectors and matrices to express geometric operations. A vector can represent a point position, a direction, an offset, or an attribute depending on context. A matrix expresses a linear map or, with homogeneous coordinates, an affine transformation.

The central exam habit is to name the coordinate space. A point in model coordinates and a light direction in view coordinates cannot be mixed safely. Many visual bugs are coordinate-space bugs.

Mental model

A numeric vector is incomplete until you know what coordinate system it belongs to.

Check yourself

Why is a normal vector not transformed exactly like a point under all transformations?

03.2 Affine Transformations

Source location: Section 3.2

Commentary

Affine transformations include translation, rotation, scaling, shearing, and combinations of these. Homogeneous coordinates allow translation to be represented in matrix form together with other transformations.

Translation moves positions but not pure directions. Rotation changes orientation while preserving lengths if it is a proper rotation. Scaling can change lengths and can also affect normals. These distinctions matter later in lighting.

Mental model

Use 4D homogeneous coordinates so that a single matrix pipeline can move 3D points through the scene.

Check yourself

What does the homogeneous w component let a transformation matrix express?

03.3 Composition of Transformations

Source location: Section 3.3

Commentary

Composition means applying several transformations in sequence. The order matters because matrix multiplication is generally not commutative. Rotating an object around its own origin and then translating it is different from translating it and then rotating around the world origin.

When you read a formula such as $\text{projection} * \text{view} * \text{model} * \text{position}$, read it from right to left for the point: first model, then view, then projection. The resulting matrix chain is compact, but the conceptual steps remain separate.

Mental model

Matrix chains are compressed stories of movement through coordinate spaces.

Check yourself

Why can swapping model and view matrices destroy the intended camera/object relation?

03.4 Coordinate System Change

Source location: Section 3.4

Commentary

A transformation can be read actively as moving an object, or passively as changing the coordinate frame used to describe it. Both interpretations are valid, but mixing them carelessly causes sign and order errors.

The view matrix is a classic example: conceptually you position a camera, but the pipeline usually transforms the world by the inverse of the camera transform so that the camera becomes the origin of view space.

Mental model

Moving the camera one way is equivalent to moving the world the opposite way for rendering.

Check yourself

Why is the view transform often related to the inverse camera transform?

03.5 Transformations in OpenGL

Source location: Section 3.5

Commentary

Modern OpenGL does not automatically manage the old matrix stacks. You usually compute model, view, and projection matrices in application code and pass them to shaders as uniforms.

The vertex shader is where the final clip-space position is normally computed. That means a wrong matrix uniform, wrong multiplication order, or wrong convention can make geometry vanish even though buffers and draw calls are correct.

Mental model

The shader is the place where abstract transformation math becomes GPU execution.

Check yourself

Which matrix would you inspect first if an object follows the camera instead of staying in the world?

End-of-lecture summary

If you remember only one thing from Lecture 03, remember this: This lecture explains how geometry moves between coordinate systems. Almost every later topic assumes that you can track where a point or vector currently lives.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.